JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to ADITYA TRADERS IOC DEALERS NH-7

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

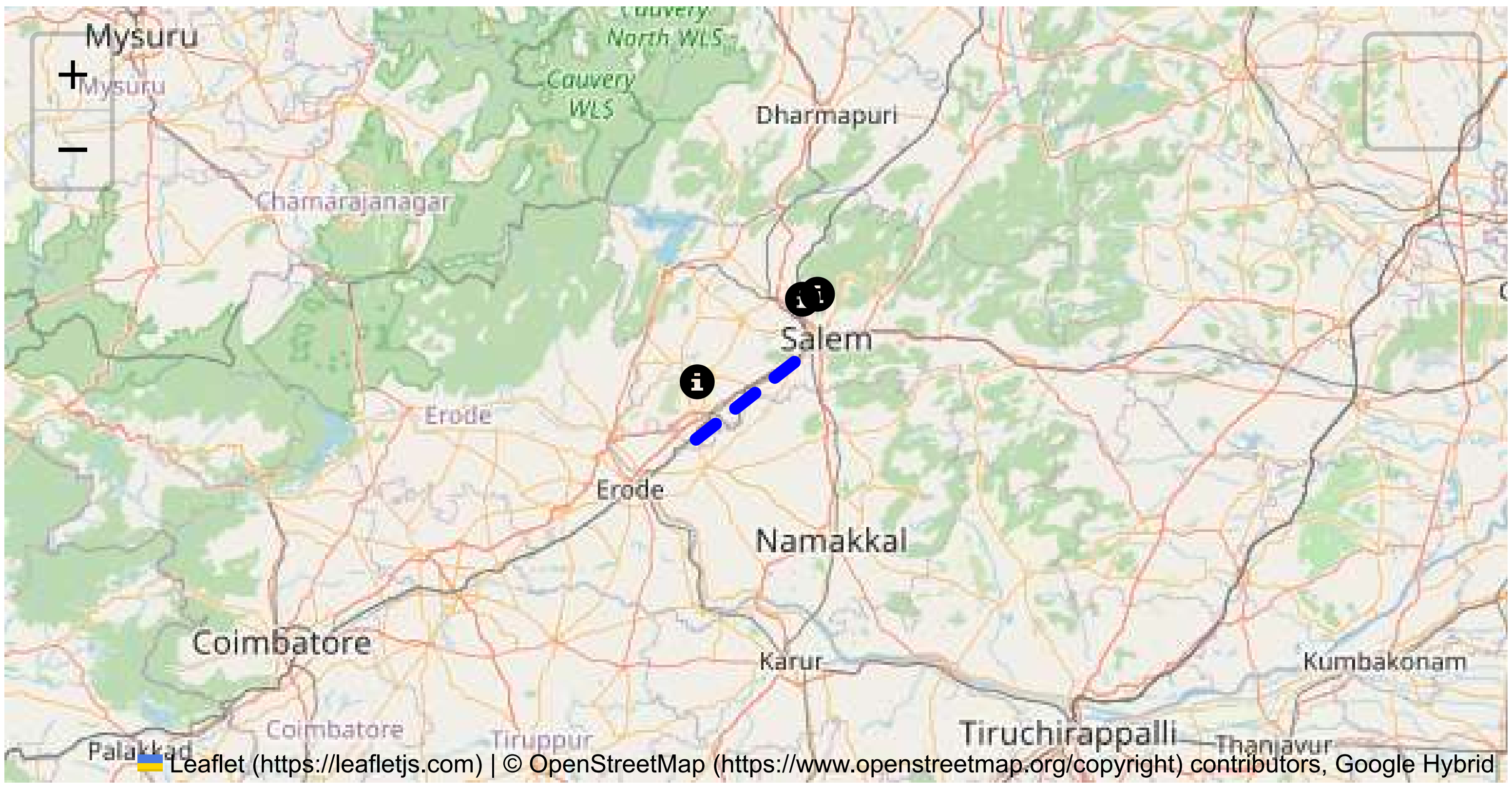
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 39.77 km
Estimated Duration: 0.9 hours
Adjusted Duration (Heavy Vehicle): 1.1 hours
Start: (11.4381, 77.8734)
End: (11.628011, 78.136253)

Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to 5/184, Seelanaikepatty Bypass Road, Maniyanur via Puthur Agraharam encompasses approximately 39.77 kilometers and typically takes about 52 minutes for trucks carrying hazardous materials. The path primarily follows state highways with segments passing through suburban and semi-rural landscapes.

1. Overview of the Route Map

The route largely involves navigating state highways with potential detours through rural areas. It forms a straightforward corridor but involves a deviation through Puthur Agraharam before reaching the destination in Maniyanur. Major intersecting roads necessitate alertness for traffic merging and diverging.

2. Typical Weather Conditions and Hazards

Tamil Nadu experiences a tropical climate. During the monsoon season (June to September), heavy rainfall can lead to waterlogged roads and reduced visibility, heightening the risk of hydroplaning and accidents. During peak summer months, high temperatures may lead to road deterioration.

3. Traffic Patterns and Congestion

Traffic patterns tend to peak during morning (8–10 AM) and evening (5–8 PM) hours, especially around urban centers. Salem is a significant urban hub, and congestion is notable around Puthur Agraharam and Seelanaikepatty.

4. Road Quality and Infrastructure

Road infrastructure varies, with well-maintained highways interspersed with narrower rural roads near Puthur Agraharam. Pay attention to potholes and uneven surfaces, particularly in rural sections and during rainy seasons when road quality can further degrade.

5. Alternative Routes

For emergencies, consider state highways linking through adjacent towns such as Sankari or detouring via major arteries like the Salem city bypass to avoid congested zones. Ensure route familiarity for quick decision-making.

6. Local Regulations on Hazardous Material Transport

Tamil Nadu regulations require hazardous material transport to adhere to safe driving standards, restricted hours, and routes. Comply with signage and avoid restricted urban zones during peak hours to prevent legal complications.

7. Historical Incidents

Historical data indicates that the majority of incidents involving heavy vehicles occur due to weather conditions and driver fatigue. There are no uncommon incidents specific to hazardous materials on this route, but standard precautions remain essential.

8. Environmental Considerations

The route passes through agricultural zones with periodic wildlife crossings. Be mindful of protection measures to avoid spillage in these sensitive areas. Implement spill containment strategies where applicable.

9. Communication Coverage

While urban areas have robust cellular coverage, isolated rural sections may experience weak signals or dead zones. employing communication equipment for emergencies ensures continuous contact with dispatch centers.

10. Estimated Emergency Response Times

Emergency response in urban fringes such as Salem is typically swift, within 15–30 minutes. In rural areas, response may extend to 45 minutes or longer. Identify the nearest health facilities and fire stations beforehand.

12. Overall Risk Assessment Summary

Overall, this route poses moderate risks characteristic of rural and urban transition corridors. Enhanced risks are predominantly weather-related and arise from infrastructure inconsistencies. Proactive route assessments and weather-aware strategies are recommended. Ensuring thorough adherence to regulations and vehicle preparedness significantly mitigates these risks, providing a safe passage for hazardous transports.

Frequent traffic updates and weather forecasts are advisable for real-time adjustments to the travel plan, ensuring a safer journey for truck drivers and goods alike.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	High	11.45007, 77.87201	15 KM/Hr
6	Turn	High	11.47407, 77.86839	15 KM/Hr
7	Turn	Medium	11.48366, 77.88781	30 KM/Hr
8	Turn	High	11.49149, 77.89601	15 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	High	11.61233, 78.10397	15 KM/Hr
11	Turn	High	11.61216, 78.10423	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	Medium	11.61284, 78.10600	30 KM/Hr
13	Turn	Medium	11.61292, 78.10606	30 KM/Hr
14	Turn	Medium	11.63306, 78.12728	30 KM/Hr
15	Turn	Medium	11.63302, 78.12742	30 KM/Hr
16	Turn	Medium	11.63292, 78.12753	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
13	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
14	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
15	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
18	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
22	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
23	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
25	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
26	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
16	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
17	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
19	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
20	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
21	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
24	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45007, 77.87201

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47407, 77.86839

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48366, 77.88781

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49149, 77.89601

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49224, 77.89601

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.61233, 78.10397

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.61216, 78.10423

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61284, 78.10600

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61292, 78.10606

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63306, 78.12728

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63302, 78.12742

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63292, 78.12753